

JOSE LUIS GUERRA

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SUMMARY

Ph.D. candidate (expected May 2026) in theoretical/computational chemistry developing multiscale, atomistic models of P-type ATPases. Expert in MD (enhanced sampling, free energy), QM/MM for chemical steps, and reproducible workflows. Built a 40+ microstate model of the Na^+, K^+ -ATPase to compute step-wise displacement charge and compare directly with available experimental measurements. Recent AI/ML internship applying computational neuroscience to create a synthetic olfactory engine. Seeking roles where I can leverage my knowledge of chemistry and computational skills to improve research. I most recently completed coursework on agentic systems for science.

TECHNICAL SKILLS

- **Programming:** Python, Bash, R, C, Fortran
- **ML/Analysis:** NumPy, pandas, SciPy, scikit-learn, PyTorch; data curation; model evaluation
- **Simulation:** NAMD, AMBER, OpenMM, Gaussian, TeraChem; enhanced sampling & free-energy methods;
- **HPC/Workflow:** Linux clusters; PBS/Slurm
- **Domain:** Ion transport, electrostatics, protein–ligand interactions, structure-based modeling

EDUCATION

University of Chicago, Chicago, IL

Ph.D. in Chemistry

Expected May 2026

NSF Graduate Research Fellow

Dissertation: Multiscale Modeling of P-type ATPase Pumps

Relevant Coursework: Mathematical Foundations of Machine Learning; Machine Learning in Chemistry, Agents for Science

Stony Brook University, Stony Brook, NY

B.S. in Biochemistry (Honors), Minor in Chemistry & Music

May 2021

Honors Thesis: Molecular Dynamics of Mismatched Base Pairs in a B-family DNA Polymerase

RESEARCH & PROFESSIONAL EXPERIENCE

Aurat Aroma LLC

AI/ML Intern (Independent Contractor)

Remote

Jul 2025 – Sep 2025

- Supported the *Synthetic Olfactory Engine* by assisting with computational neuroscience models and supporting algorithms for future integration.
- Developed production-quality Python code and technical documentation for new components, ensuring clear IP boundaries (no reuse of prior engagements).
- Contributed to back-end infrastructure and data pipelines aligned with scientific and technical objectives; collaborated with the founder and R&D to translate requirements into implementable tasks.

Roux Lab, University of Chicago

Graduate Researcher

Chicago, IL

Jan 2022 – Present

- Built an atomistic, 40+ microstate model of the Na⁺,K⁺-ATPase transport cycle; designed analyses to compute step-wise displacement charge and link mechanism to measurable observables; reproduced available measurements to interpret ion binding/occlusion.
- Designed enhanced-sampling and free-energy workflows integrated with QM/MM to study ATP hydrolysis / dephosphorylation; built reproducible Python pipelines that processed and validated >1 TB of simulation data.

Amgen Scholars Program, Columbia University

Summer Research Intern

New York, NY

Summer 2021

- Established protocols for high-speed AFM imaging of biomechanical properties.

Simmerling Lab, Stony Brook University

Undergraduate Researcher

Stony Brook, NY

Jan 2018 – May 2021

- Simulated protein–nucleic acid interactions using AMBER/NAMD and Python (NumPy, pandas); validated results via statistical analyses and targeted *in silico* mutations.

LEADERSHIP & SERVICE

- **UChicago PSD Office of Diversity and Inclusion:** Graduate Student Intern *Dec 2023–May 2025*: Developed a web-scraping pipeline to gather and analyze demographic data from R1 institutions; evaluated existing partnerships with R1 and R2 minority-serving institutions to inform targeted recruitment and outreach strategies.
- **Mentoring:** Graduate school application mentor through Científico Latino (2021–2024): provided resume and cover letter feedback, conducted mock interviews, and supported students from underrepresented backgrounds applying to STEM Ph.D. programs.

SELECTED PUBLICATIONS & PRESENTATIONS

Poster: *Multiscale Modeling of P-type ATPase Pumps*. Gateway Ion Channel Symposium, July 2026.

Guerra, J., and Roux, B. (2025). “Multiscale Modeling of P-type Pumps.” *In preparation*.

Guerra, J., Rui, H., and Roux, B. (2025). “Multistate Kinetic Model of the Sodium–Potassium ATPase.” *J. Phys. Chem. B*.

Andrews, B., **Guerra, J.**, et al. (2021). “Do Molecular Dynamics Force Fields Accurately Model Ramachandran Distributions of Amino Acid Residues in Water?” *Phys. Chem. Chem. Phys.*

Fallon, L., et al. (2021). “Free Energy Landscapes from SARS-CoV-2 Spike Glycoprotein Simulations Suggest that RBD Opening Can Be Modulated via Interactions in an Allosteric Pocket.” *J. Am. Chem. Soc.*

AWARDS & HONORS

National Science Foundation Graduate Research Fellowship

University of Chicago McCormick Graduate Fellowship

Protein Society Diversity Award

Provost Award for Academic Excellence